

Stack, Queue & Priority Queue Templates

Four data structures — each with a canonical template and representative problems.

Quick Reference Table

#	Name	Description	Intuition	Time	Space	Key Implementation
20	Valid Parentheses	Given a string of brackets <code>()[]{} </code> , return true if all brackets are properly closed and ordered.	the most recent unmatched open bracket is the only one a closing bracket can legally match — that is exactly LIFO.	$O(n)$	$O(n)$	Standard
155	Min Stack	Design a stack that supports push, pop, top, and <code>getMin()</code> in $O(1)$.		$O(1)$ per operation	$O(n)$	Standard
394	Decode String	Decode a string like <code>"3[a2[bc]]"</code> → <code>"abcbcabcbc"</code> .		$O(n)$ (n = length of decoded output)	$O(n)$	Standard
739	Daily Temperatures	For each day, how many days until a warmer temperature? Return 0 if none.	a colder day just waits on the stack until the first warmer day arrives and resolves it.	$O(n)$	$O(n)$	Standard
496	Next Greater Element I	For each element in <code>nums1</code> (a subset of <code>nums2</code>), find the first greater element to its right in <code>nums2</code> . Return -1 if none.		$O(m + n)$	$O(n)$	Standard
84	Largest Rectangle in Histogram	Find the largest rectangle that can be formed within the histogram bars.	a bar can only stretch sideways until it hits a shorter bar; the stack remembers each bar's left limit so the right limit (current shorter bar) closes the rectangle.	$O(n)$	$O(n)$	Standard
42	Trapping Rain Water	How much water can be trapped between the bars after rain?	water sits in a dip between a left wall and a right wall, and its depth is set by whichever wall is shorter.	$O(n)$	$O(n)$	Standard
239	Sliding Window Maximum	Return the maximum in each sliding window of size k .	any element smaller than a newer element can never be the window max again, so discard it; the front always holds the current best.	$O(n)$	$O(k)$	Standard
1046	Last Stone Weight	Smash the two heaviest stones repeatedly. Return the last remaining weight (or 0).		$O(n \log n)$	$O(n)$	Standard
347	Top K Frequent Elements	Return the k most frequent elements.		$O(n \log k)$	$O(n)$	Standard
295	Find Median from Data Stream	Add integers to a stream and find the median at any time in $O(\log n)$ add and $O(1)$ find.	keep the smaller half and larger half balanced; the median always lives right at the boundary, on the tops of the two heaps.	$O(\log n)$ addNum, $O(1)$ findMedian	$O(n)$	Standard
502	IPO	Before each project you must have enough capital. Start with capital <code>w</code> . Pick at most <code>k</code> projects to maximize final capital.		$O(n \log n)$	$O(n)$	Standard
767	Reorganize String	Rearrange characters so no two adjacent characters are the same. Return <code>""</code> if impossible.		$O(n \log k)$ (k = distinct chars ≤ 26)	$O(k)$	Standard
503	Next Greater Element II	Given a circular integer array, return the next greater number for every element. The next greater number of an element is the first greater number found by traversing the array circularly.		$O(n)$	$O(n)$	Monotone decreasing stack. Iterate through the array TWICE (indices 0 to $2n-1$, using <code>i % n</code>). Only push index <code>i</code> onto the stack during the first pass (<code>i < n</code>) to avoid duplicates.
85	Maximal Rectangle	Given a binary matrix filled with 0s and 1s, find the largest rectangle containing only 1s and return its area.		$O(m \cdot n)$	$O(n)$	For each row, compute cumulative histogram heights (1s stack vertically). Then apply the Largest Rectangle in Histogram algorithm (#84) on each row's histogram.
224	Basic Calculator	Evaluate a string expression containing <code>+</code> , <code>-</code> , <code>(</code> , <code>)</code> , digits, and spaces.		$O(n)$	$O(n)$	When encountering <code>(</code> , push current (result, sign) onto stack and reset. When encountering <code>)</code> , combine current result with the saved result and sign from the stack.
227	Basic Calculator II	Evaluate a string expression with <code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> and integers (no parentheses).		$O(n)$	$O(n)$	Process operator BEFORE the current number. Push positive/negative numbers for <code>+</code> / <code>-</code> . For <code>*</code> / <code>/</code> , immediately multiply/divide the top of the stack. Sum the stack at the end.

#	Name	Description	Intuition	Time	Space	Key Implementation
3 7 8	Kth Smallest Element in a Sorted Matrix	Given an $n \times n$ matrix where each row and column is sorted in ascending order, return the k th smallest element.		$O(n \log(\max - \min))$	$O(1)$	Binary search on the VALUE range <code>[matrix[0][0], matrix[n-1][n-1]]</code> . For a given <code>mid</code> , count elements $\leq$ <code>mid</code> by scanning from the bottom-left corner: if <code>matrix[row][col] <= mid</code> , all <code>row+1</code> elements in this column (<code>0..row</code>) are $\leq$ <code>mid</code> .
3 7 3	Find K Pairs with Smallest Sums	Given two sorted arrays <code>nums1</code> and <code>nums2</code> , return the <code>k</code> pairs <code>(u, v)</code> (one from each) with the smallest sums.		$O(k \log k)$	$O(k)$	Seed min-heap with <code>(nums1[i], nums2[0])</code> for $i = 0..\min(k, n1.length)-1$. On each pop, if <code>j+1 < nums2.length</code> , push <code>(nums1[i], nums2[j+1])</code> . This expands row-by-row in the implicit pair matrix.
6 2 1	Task Scheduler	Given a list of tasks (characters) and a cooldown <code>n</code> , return the minimum number of intervals to finish all tasks. Between two same tasks there must be at least <code>n</code> intervals.		$O(k)$	$O(1)$	Greedy formula. Find the most frequent task (<code>maxFreq</code>). It creates <code>maxFreq - 1</code> "frames" of <code>n + 1</code> slots, plus <code>maxCount</code> (count of tasks with <code>maxFreq</code> slots. The answer is <code>max(tasks.length, (maxFreq - 1) * (n + 1) + maxCount)</code> .
3 5 8	Rearrange String k Distance Apart	Rearrange a string so that the same characters are at least <code>k</code> distance apart. Return "" if impossible.		$O(n \log k)$	$O(k)$	greedy with a max-heap by frequency plus a cooldown queue of size <code>k</code> that holds recently used characters until they're eligible again.
4 8 0	Sliding Window Median	Return the median of every window of size <code>k</code> as it slides across the array.		$O(n-k)$ with heap remove ($O(n \log k)$ with indexed sets)	$O(k)$	two heaps (<code>maxHeap</code> = lower half, <code>minHeap</code> = upper half) with lazy deletion — defer removing elements that left the window, tracking balance with counts.
6 9 2	Top K Frequent Words	Return the <code>k</code> most frequent words. Sort by frequency descending; ties broken by lexicographic order.		$O(n \log k)$	$O(n)$	min-heap of size <code>k</code> ordered so the "smallest" (lowest freq, or same freq with lexicographically larger word) sits on top for eviction.
7 7 2	Basic Calculator III	Evaluate an arithmetic expression with <code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> , and parentheses.		$O(n)$	$O(n)$	combines #224 (parentheses via recursion) and #227 (operator precedence: apply <code>*</code> / <code>/</code> immediately, defer <code>+</code> / <code>-</code>).
3 2	Longest Valid Parentheses	Find the length of the longest valid (well-formed) parentheses substring.	push indices onto a stack; the bottom of the stack is always the index just before the current valid run, so the gap to it gives the run length.	$O(n)$	$O(n)$	Standard
7 1	Simplify Path	Simplify an absolute Unix file path (handle <code>.</code> , <code>..</code> , multiple slashes).	split on <code>/</code> and treat directories as a stack — <code>..</code> pops the last directory, <code>.</code> and empty tokens are ignored.	$O(n)$	$O(n)$	Standard
1 5 0	Evaluate Reverse Polish Notation	Evaluate a Reverse Polish Notation expression with <code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> .	push operands; on an operator pop the two most recent operands, apply, and push the result back — exactly LIFO.	$O(n)$	$O(n)$	Standard
2 1 5	Kth Largest Element in an Array	Find the k th largest element in an unsorted array (not necessarily distinct).	a min-heap of size <code>k</code> keeps the k largest seen so far; its top is the k th largest.	$O(n \log k)$	$O(k)$	Standard
2 1 8	The Skyline Problem	Given building rectangles, return the skyline as a list of key points.	sweep left to right over building edges; a max-heap of active heights tracks the tallest standing building, and a key point is emitted whenever that maximum changes.	$O(n^2)$ (heap remove)	$O(n)$	Standard
2 2 5	Implement Stack using Queues	Implement a LIFO stack using only queue operations.	after each push, rotate the queue so the newest element sits at the front — then <code>poll</code> / <code>peek</code> behave like <code>stack pop</code> / <code>top</code> .	$O(n)$ push, $O(1)$ others	$O(n)$	Standard
2 3 2	Implement Queue using Stacks	Implement a queue using two stacks.	one stack receives pushes, the other serves pops; moving elements over reverses their order, restoring FIFO. Amortized $O(1)$ per element.	$O(1)$ amortized	$O(n)$	two stacks emulate FIFO
3 1 6	Remove Duplicate Letters	Remove duplicate letters so each appears once, result is lexicographically smallest.	monotone increasing stack of letters — pop a larger letter when a smaller one arrives, but only if the popped letter appears again later (so we can re-add it).	$O(n)$	$O(1)$ (26 letters)	Standard
4 0 2	Remove K Digits	Remove <code>k</code> digits from a number string to get the smallest possible number.	monotone increasing stack of digits — whenever a smaller digit arrives, pop larger preceding digits (each pop is one removal) so high places hold the smallest digits.	$O(n)$	$O(n)$	Standard

#	Name	Description	Intuition	Time	Space	Key Implementation
4 5 6	132 Pattern	Check if a 1-3-2 pattern exists in an integer array.		$O(n)$	$O(n)$	scan right to left with a monotone decreasing stack; pop to track the largest value that is still smaller than some element to its right (the "2"). If a later element is below that "2", a 132 pattern exists.
5 0 6	Relative Ranks	Assign ranks ("Gold Medal", "Silver Medal", "Bronze Medal", then 4, 5, ...) to athletes by score.	a max-heap of (score, index) pops athletes in descending score order, so each pop assigns the next rank back to its original position.	$O(n \log n)$	$O(n)$	Standard
6 3 6	Exclusive Time of Functions	Given a log of function start/end times, compute exclusive time per function.	a call stack mirrors execution; when a new call starts, the currently running function pauses (accumulate its slice), and when a call ends, the resumed parent restarts its clock.	$O(L)$ (L = number of logs)	$O(n)$	Standard
6 7 8	Valid Parenthesis String	Check if a string with (,) , * (wildcard) can be a valid parenthesis string.		$O(n)$	$O(1)$	track a range [low, high] of possible open-paren counts; * widens the range (could be (,) , or empty). Valid iff the range can return to 0 and never goes negative.
7 0 3	Kth Largest Element in a Stream	Design a class to find the kth largest element in a stream.	a min-heap capped at size k always holds the k largest seen; its top is the running kth largest.	$O(\log k)$ per add	$O(k)$	Standard
7 1 6	Max Stack	Design a stack with push, pop, top, getMax, and popMax operations.		$O(n)$ popMax, $O(1)$ others	$O(n)$	mirror an auxiliary maxStack (like #155). popMax pops down to the max into a temp buffer, removes it, then pushes the buffer back — re-establishing both stacks.
7 3 5	Asteroid Collision	Simulate asteroids moving in a row: right-moving collide with left-moving.	a stack holds surviving asteroids; a left-moving asteroid only collides with a right-moving one on top, resolving collisions repeatedly until it survives, explodes, or the stack clears.	$O(n)$	$O(n)$	Standard
7 5 9	Employee Free Time	Find the free time intervals common to all employees given their schedules.	flatten all intervals, sort by start, then sweep tracking the furthest end seen so far; any gap between that end and the next start is common free time.	$O(n \log n)$	$O(n)$	Standard
8 4 4	Backspace String Compare	Check if two strings are equal after processing # as backspace.	build each string with a stack where # pops the last character, then compare the resulting stacks.	$O(n)$	$O(n)$	Standard
8 5 6	Score of Parentheses	Calculate the score of a valid parentheses string (empty=0, AB=A+B, (A)=2A or 1 if empty).		$O(n)$	$O(n)$	stack holds the accumulated score at each depth; push 0 on (, and on) collapse the inner score (max(2*inner, 1)) into the parent frame.
8 6 2	Shortest Subarray with Sum at Least K	Find the length of the shortest subarray with sum at least k (k can be negative).		$O(n)$	$O(n)$	monotone increasing deque over prefix sums — pop from the front when a window qualifies, and from the back when a newer prefix is smaller (dominating older larger ones).
9 0 7	Sum of Subarray Minimums	Find the sum of subarray minimums for all subarrays.		$O(n)$	$O(n)$	monotone increasing stack — for each element as the minimum, count subarrays via distance to the previous strictly-smaller and next smaller-or-equal element, then weight by the value.
9 2 1	Minimum Add to Make Parentheses Valid	Find the minimum additions to make a parentheses string valid.	track open parentheses as a counter (stack of size); each unmatched) needs an added (, and leftover (each need a) .	$O(n)$	$O(1)$	Standard
9 4 6	Validate Stack Sequences	Check if a sequence can be produced by a series of push-pop operations on a stack.	simulate — push each value, then greedily pop whenever the stack top matches the next expected popped value. If everything pops, the sequence is valid.	$O(n)$	$O(n)$	Standard
9 7 3	K Closest Points to Origin	Find the k points closest to the origin.	a max-heap of size k keyed on squared distance keeps the k closest seen; evict the farthest whenever the heap overflows.	$O(n \log k)$	$O(k)$	Standard
1 0 8 6	High Five	For each student, compute the average of their top five scores.	keep a min-heap of size 5 per student so only their five highest scores remain, then average those.	$O(n \log 5)$	$O(n)$	Standard
1 1 9 0	Reverse Substrings Between	Reverse the substrings between each pair of parentheses (innermost first).		$O(n^2)$	$O(n)$	stack of builders — push current builder on (; on) reverse the inner builder and append it to the parent frame.

#	Name	Description	Intuition	Time	Space	Key Implementation
	Each Pair of Parentheses					
1 2 0 9	Remove All Adjacent Duplicates in String II	Remove all adjacent duplicates of length k in a string repeatedly.		$O(n)$	$O(n)$	stack of (char, count) pairs — increment the top's count on a repeat, and pop the frame once its count reaches k.
1 2 4 9	Minimum Remove to Make Valid Parentheses	Remove the minimum number of parentheses to make the string valid.		$O(n)$	$O(n)$	stack stores indices of unmatched (; a) with no match is marked for removal, and any (left on the stack at the end is also removed.
1 3 3 7	The K Weakest Rows in a Matrix	Find the k weakest rows in a matrix (fewest soldiers, ties broken by row index).		$O(m \cdot n + m \log k)$	$O(k)$	max-heap of size k keyed on (soldier count, row index) so the strongest qualifying row sits on top for eviction; the remaining k are the weakest.
1 5 4 1	Minimum Insertions to Balance a Parentheses String	Find the minimum insertions to make a string balanced (every (needs two)).		$O(n)$	$O(1)$	counter-style stack tracking open (; each (demands two) . Handle) carefully since they come in pairs, inserting a) when only one is available.
1 6 1 4	Maximum Nesting Depth of the Parentheses	Find the maximum nesting depth of parentheses in a string.	the stack depth equals the current nesting level; track a running counter for (/) and record its peak.	$O(n)$	$O(1)$	Standard
1 7 9 2	Maximum Average Pass Ratio	Maximize the average pass ratio by adding extra students optimally.		$O((n + \text{extra}) \log n)$	$O(n)$	max-heap keyed on the marginal gain from adding one student to a class; greedily assign each extra student where it helps most, then push the updated gain back.
1 9 4 4	Number of Visible People in a Queue	Count the number of people each person can see in a queue (taller blocks view).		$O(n)$	$O(n)$	monotone decreasing stack scanned right to left; pop shorter people (each is visible) until a taller one blocks the view — that taller person is visible too.
1 9 8 5	Find the Kth Largest Integer in the Array	Find the kth largest integer in an array of number strings.		$O(n \log k)$	$O(k)$	min-heap of size k comparing the numeric strings by length first, then lexicographically (so big-integer values compare correctly without overflow).

All Templates

```
// 1. STACK (LIFO) – use ArrayDeque, not Stack class
Deque<Integer> stack = new ArrayDeque<>();
stack.push(x);    // add to top
stack.pop();      // remove from top
stack.peek();     // view top, no remove

// 2A. MONOTONE DECREASING STACK – next GREATER element
// MENTAL MODEL: keep only candidates still waiting for a bigger neighbor; current resolves them all.
// WHEN: "next/previous greater element", "warmer/taller to the right"
// Invariant: stack values decrease from bottom to top
// Pop when current is GREATER than top → top found its next greater
Deque<Integer> stack = new ArrayDeque<>(); // stores indices
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] < nums[i])
        result[stack.pop()] = nums[i];    // nums[i] is next greater for popped index
    stack.push(i);
}
// Remaining in stack: no next greater element exists

// 2B. MONOTONE INCREASING STACK – next SMALLER element / span width
// MENTAL MODEL: each popped bar's reach extends until something shorter blocks it on both sides.
// WHEN: "largest rectangle", "trapped water", "span/width bounded by smaller elements"
// Invariant: stack values increase from bottom to top
// Pop when current is SMALLER than top → use popped index to compute width/span
Deque<Integer> stack = new ArrayDeque<>(); // stores indices
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] > nums[i]) {
        int height = nums[stack.pop()];
        int width = stack.isEmpty() ? i : i - stack.peek() - 1; // span between boundaries
        // process height * width
    }
    stack.push(i);
}

// 3. MONOTONE DEQUE – sliding window max/min
// MENTAL MODEL: the front is the current window's answer; anything a newer-and-bigger value beats is dead weight.
// WHEN: "max/min of every window of size k"
// Deque stores INDICES; values at those indices are decreasing front-to-back (for max)
Deque<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!queue.isEmpty() && nums[queue.peekLast()] <= nums[i])
        queue.pollLast();    // remove smaller elements – they can never be window max
    queue.offerLast(i);
    if (queue.peekFirst() < i - k + 1) queue.pollFirst(); // evict out-of-window elements
    if (i >= k - 1) result[i - k + 1] = nums[queue.peekFirst()];
}

// 4. PRIORITY QUEUE
PriorityQueue<Integer> minPQ = new PriorityQueue<>();    // min-heap (default)
PriorityQueue<Integer> maxPQ = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // custom: sort by index 1
pq.offer(x);    // add
pq.poll();     // remove and return min/max
pq.peek();     // view min/max without removing

// 5. TWO HEAPS – maintain median in O(log n)
// MENTAL MODEL: split the data at the median; the median is always sitting on the two heaps' tops.
// WHEN: "running median", "median of a data stream"
// maxHeap = lower half (smaller numbers), minHeap = upper half (larger numbers)
// Invariant: maxHeap.size() == minHeap.size() or maxHeap.size() == minHeap.size() + 1
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
void addNum(int num) {
```

```

maxHeap.offer(num);
minHeap.offer(maxHeap.poll());           // push one to minHeap (possibly num itself)
if (maxHeap.size() < minHeap.size())
    maxHeap.offer(minHeap.poll());       // rebalance: maxHeap always ≥ minHeap in size
}
double findMedian() {
    return maxHeap.size() > minHeap.size()
        ? maxHeap.peek()
        : (maxHeap.peek() + minHeap.peek()) / 2.0;
}

```

Part 1 — Basic Stack

#20 Valid Parentheses

Description: Given a string of brackets `()[]{}` , return true if all brackets are properly closed and ordered.

Intuition: the most recent unmatched open bracket is the only one a closing bracket can legally match — that is exactly LIFO.

```

class Solution {
    public boolean isValid(String s) {
        Deque<Character> stack = new ArrayDeque<>();
        for (char c : s.toCharArray()) {
            if (c == '(' || c == '[' || c == '{') {
                stack.push(c);
            } else {
                if (stack.isEmpty()) return false;
                char top = stack.pop();
                if (c == ')' && top != '(') return false;
                if (c == ']' && top != '[') return false;
                if (c == '}' && top != '{') return false;
            }
        }
        return stack.isEmpty();
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#155 Min Stack

Description: Design a stack that supports push, pop, top, and `getMin()` in $O(1)$.

Key: auxiliary `minStack` mirrors the main stack — each level stores the current running minimum.

```

class MinStack {
    Deque<Integer> stack    = new ArrayDeque<>();
    Deque<Integer> minStack = new ArrayDeque<>(); // ← always push current min alongside value

    public void push(int val) {
        stack.push(val);
        minStack.push(minStack.isEmpty() ? val : Math.min(minStack.peek(), val));
    }
    public void pop()    { stack.pop(); minStack.pop(); } // ← pop both in sync
    public int top()     { return stack.peek(); }
    public int getMin()  { return minStack.peek(); }
}

```

Time $O(1)$ per operation | **Space** $O(n)$

#394 Decode String

Description: Decode a string like "3[a2[bc]]" → "abcbcabcbcb".

Key: two stacks — one for repeat counts, one for the string built before each [. On] , pop both and repeat.

```
class Solution {
    public String decodeString(String s) {
        Deque<Integer> countStack = new ArrayDeque<>();
        Deque<StringBuilder> strStack = new ArrayDeque<>();
        StringBuilder current = new StringBuilder();
        int k = 0;
        for (char c : s.toCharArray()) {
            if (Character.isDigit(c)) {
                k = k * 10 + (c - '0'); // ← multi-digit number
            } else if (c == '[') {
                countStack.push(k); // ← save count
                strStack.push(current); // ← save string so far
                current = new StringBuilder();
                k = 0;
            } else if (c == ']') {
                int count = countStack.pop();
                StringBuilder parent = strStack.pop();
                for (int i = 0; i < count; i++) parent.append(current); // ← repeat
                current = parent;
            } else {
                current.append(c);
            }
        }
        return current.toString();
    }
}
```

Time O(n) (n = length of decoded output) | **Space** O(n)

Part 2 — Monotone Stack

Decreasing vs Increasing

DECREASING stack (pop when current > top):	finds NEXT GREATER element for each popped index
INCREASING stack (pop when current < top):	finds NEXT SMALLER element — used for span width

Both store INDICES (not values) so you can compute distances and widths.

#739 Daily Temperatures

Description: For each day, how many days until a warmer temperature? Return 0 if none.

Template: monotone decreasing — when current temp is greater, pop and record the gap.

Intuition: a colder day just waits on the stack until the first warmer day arrives and resolves it.

```

class Solution {
    public int[] dailyTemperatures(int[] temperatures) {
        int n = temperatures.length;
        int[] result = new int[n];
        Deque<Integer> stack = new ArrayDeque<>(); // indices; temps decrease from bottom to top
        for (int i = 0; i < n; i++) {
            while (!stack.isEmpty() && temperatures[stack.peek()] < temperatures[i]) {
                int idx = stack.pop();
                result[idx] = i - idx; // ← days until warmer = distance
            }
            stack.push(i);
        }
        return result;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#496 Next Greater Element I

Description: For each element in `nums1` (a subset of `nums2`), find the first greater element to its right in `nums2`. Return -1 if none.

Key: precompute NGE for all elements in `nums2` using a monotone stack + HashMap. Then look up each `nums1` element.

```

class Solution {
    public int[] nextGreaterElement(int[] nums1, int[] nums2) {
        Map<Integer, Integer> nextGreater = new HashMap<>();
        Deque<Integer> stack = new ArrayDeque<>();
        for (int num : nums2) {
            while (!stack.isEmpty() && stack.peek() < num)
                nextGreater.put(stack.pop(), num); // ← num is next greater for stack.peek()
            stack.push(num);
        }
        int[] result = new int[nums1.length];
        for (int i = 0; i < nums1.length; i++)
            result[i] = nextGreater.getOrDefault(nums1[i], -1);
        return result;
    }
}

```

Time $O(m + n)$ | **Space** $O(n)$

#84 Largest Rectangle in Histogram

Description: Find the largest rectangle that can be formed within the histogram bars.

Template: monotone increasing stack (pop when current bar is shorter). Popped bar's width = from `stack.peek() + 1` to `i - 1`.

Key: add sentinel `0`'s at both ends to flush all remaining bars from the stack.

Intuition: a bar can only stretch sideways until it hits a shorter bar; the stack remembers each bar's left limit so the right limit (current shorter bar) closes the rectangle.

```

class Solution {
    public int largestRectangleArea(int[] heights) {
        int n = heights.length;
        int[] h = new int[n + 2];           // ← sentinels: h[0]=h[n+1]=0
        System.arraycopy(heights, 0, h, 1, n);
        Deque<Integer> stack = new ArrayDeque<>();
        int maxArea = 0;
        for (int i = 0; i <= n + 1; i++) {
            while (!stack.isEmpty() && h[stack.peek()] > h[i]) {
                int height = h[stack.pop()];
                int left = stack.isEmpty() ? -1 : stack.peek(); // left boundary
                int width = i - left - 1;                       // bars strictly between left boundary and current
                maxArea = Math.max(maxArea, height * width);
            }
            stack.push(i);
        }
        return maxArea;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#42 Trapping Rain Water

Description: How much water can be trapped between the bars after rain?

Template: monotone increasing stack — when current bar is taller, water fills the "valley" between the current bar and the bar now at the top of the stack.

Intuition: water sits in a dip between a left wall and a right wall, and its depth is set by whichever wall is shorter.

```

class Solution {
    public int trap(int[] height) {
        Deque<Integer> stack = new ArrayDeque<>();
        int water = 0;
        for (int i = 0; i < height.length; i++) {
            while (!stack.isEmpty() && height[stack.peek()] < height[i]) {
                int bottom = height[stack.pop()];
                if (stack.isEmpty()) break;
                int left = stack.peek();
                int h = Math.min(height[left], height[i]) - bottom; // bounded by the shorter wall, minus the floor height
                int w = i - left - 1;
                water += h * w;
            }
            stack.push(i);
        }
        return water;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

84 vs 42 — Side by Side

	#84 Largest Rectangle	#42 Trapping Rain Water
Pop when:	$h[\text{current}] < h[\text{top}]$	$h[\text{current}] > h[\text{top}]$
Stack order:	increasing (bottom→top)	increasing (bottom→top)
Popped bar role:	height of rectangle	floor of the trapped water valley
Width from:	$(i - \text{stack.peek}() - 1)$	$(i - \text{stack.peek}() - 1)$
Area/water:	height × width	$\min(\text{left_wall}, \text{right_wall}) \times \text{width}$

Part 3 — Monotone Deque

#239 Sliding Window Maximum

Description: Return the maximum in each sliding window of size k.

Template: deque stores indices; values at those indices are strictly decreasing front-to-back (max at front).

Intuition: any element smaller than a newer element can never be the window max again, so discard it; the front always holds the current best.

```
class Solution {
    public int[] maxSlidingWindow(int[] nums, int k) {
        int n = nums.length;
        int[] result = new int[n - k + 1];
        Deque<Integer> queue = new ArrayDeque<>(); // indices, values decrease front-back
        for (int i = 0; i < n; i++) {
            while (!queue.isEmpty() && nums[queue.peekLast()] <= nums[i])
                queue.pollLast(); // ← remove smaller: they can never be max
            queue.offerLast(i);
            if (queue.peekFirst() < i - k + 1)
                queue.pollFirst(); // ← evict index out of window
            if (i >= k - 1)
                result[i - k + 1] = nums[queue.peekFirst()]; // ← front = window max
        }
        return result;
    }
}
```

Time $O(n)$ | **Space** $O(k)$

Part 4 — Priority Queue

#1046 Last Stone Weight

Description: Smash the two heaviest stones repeatedly. Return the last remaining weight (or 0).

Template: max-heap — always poll the two largest.

```
class Solution {
    public int lastStoneWeight(int[] stones) {
        PriorityQueue<Integer> pq = new PriorityQueue<>(Collections.reverseOrder());
        for (int stone : stones) pq.offer(stone);
        while (pq.size() > 1) {
            int a = pq.poll(), b = pq.poll();
            if (a != b) pq.offer(a - b); // ← only add remainder if different
        }
        return pq.isEmpty() ? 0 : pq.poll();
    }
}
```

Time $O(n \log n)$ | **Space** $O(n)$

#347 Top K Frequent Elements

Description: Return the k most frequent elements.

Template: min-heap of size k — evict the least frequent when size exceeds k. What remains = top k.

```

class Solution {
    public int[] topKFrequent(int[] nums, int k) {
        Map<Integer, Integer> freq = new HashMap<>();
        for (int num : nums) freq.merge(num, 1, Integer::sum);
        PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // min-heap on freq
        for (Map.Entry<Integer, Integer> e : freq.entrySet()) {
            pq.offer(new int[]{e.getKey(), e.getValue()});
            if (pq.size() > k) pq.poll(); // ← evict least frequent
        }
        return pq.stream().mapToInt(a -> a[0]).toArray();
    }
}

```

Time $O(n \log k)$ | **Space** $O(n)$

#295 Find Median from Data Stream

Description: Add integers to a stream and find the median at any time in $O(\log n)$ add and $O(1)$ find.

Template: two heaps — `maxHeap` holds lower half, `minHeap` holds upper half. Always rebalance so `maxHeap.size() ≥ minHeap.size()`.

Intuition: keep the smaller half and larger half balanced; the median always lives right at the boundary, on the tops of the two heaps.

```

class MedianFinder {
    PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder()); // lower half
    PriorityQueue<Integer> minHeap = new PriorityQueue<>(); // upper half

    public void addNum(int num) {
        maxHeap.offer(num);
        minHeap.offer(maxHeap.poll()); // ← always push one to minHeap first
        if (maxHeap.size() < minHeap.size())
            maxHeap.offer(minHeap.poll()); // ← rebalance: maxHeap always ≥ minHeap size
    }
    public double findMedian() {
        return maxHeap.size() > minHeap.size()
            ? maxHeap.peek()
            : (maxHeap.peek() + minHeap.peek()) / 2.0;
    }
}

```

Time $O(\log n)$ addNum, $O(1)$ findMedian | **Space** $O(n)$

#502 IPO

Description: Before each project you must have enough capital. Start with capital `w`. Pick at most `k` projects to maximize final capital.

Template: sort by required capital + max-heap of profits. At each step, unlock all affordable projects (move them into the heap), then greedily pick the most profitable.

```

class Solution {
    public int findMaximizedCapital(int k, int w, int[] profits, int[] capital) {
        int n = profits.length;
        int[][] projects = new int[n][2];
        for (int i = 0; i < n; i++) projects[i] = new int[]{capital[i], profits[i]};
        Arrays.sort(projects, (a, b) -> a[0] - b[0]); // ← sort by required capital
        PriorityQueue<Integer> pq = new PriorityQueue<>(Collections.reverseOrder()); // max-heap of profits
        int idx = 0;
        for (int i = 0; i < k; i++) {
            while (idx < n && projects[idx][0] <= w) pq.offer(projects[idx++][1]); // ← unlock
            if (pq.isEmpty()) break;
            w += pq.poll(); // ← pick most profitable
        }
        return w;
    }
}

```

#767 Reorganize String

Description: Rearrange characters so no two adjacent characters are the same. Return "" if impossible.
Template: max-heap of (char, count) — at each step, pick the two most frequent chars and append them alternately.

```
class Solution {
    public String reorganizeString(String s) {
        int[] count = new int[26];
        for (char c : s.toCharArray()) count[c - 'a']++;
        PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> b[1] - a[1]); // max-heap on count
        for (int i = 0; i < 26; i++) if (count[i] > 0) pq.offer(new int[]{i, count[i]});
        StringBuilder builder = new StringBuilder();
        while (pq.size() > 1) {
            int[] a = pq.poll(), b = pq.poll();
            builder.append((char)('a' + a[0]));
            builder.append((char)('a' + b[0]));
            if (--a[1] > 0) pq.offer(a);
            if (--b[1] > 0) pq.offer(b);
        }
        if (!pq.isEmpty()) {
            int[] last = pq.poll();
            if (last[1] > 1) return ""; // ← only one char left, count > 1: impossible
            builder.append((char)('a' + last[0]));
        }
        return builder.toString();
    }
}
```

Structure Chooser

Goal	Structure	Why
Next greater/smaller element	Monotone stack	$O(n)$ total — each element pushed/popped once
Sliding window max/min	Monotone deque	Front = window extreme; stale indices evicted
Global min/max, k-th element	PriorityQueue	$O(\log n)$ per op
Running median	Two heaps	Split at median; each heap half is $O(\log n)$
$O(1)$ stack minimum	Aux min-stack	Mirror min value at every level
Nested structure (brackets, k-repeats)	Two stacks	One for counts, one for strings

Deque API Cheat Sheet

One class, three roles. `ArrayDeque` adds/removes at *both* ends, so stack, queue, and deque are just usage patterns of the same structure — the only difference is **which end you remove from**. Always declare it `Deque<Integer> queue = new ArrayDeque<>();` (never the legacy `Stack` class, which is synchronized and `Vector`-backed; and `ArrayDeque` beats `LinkedList` as a queue). Name the variable for the *role* it plays (`stack` / `queue`).

addFirst / push

↓

[FRONT BACK]

↑

pollFirst / pop

addLast / offer

↓

[FRONT BACK]

↑

pollLast

STACK (LIFO): add + remove at FRONT → newest out first

QUEUE (FIFO): add at BACK, remove at FRONT → oldest out first

Intent	Stack (LIFO)	Queue (FIFO)
Add	push(x) → addFirst	offer(x) → addLast
Remove	pop() → removeFirst	poll() → removeFirst
Peek	peek() → peekFirst	peek() → peekFirst

Both **remove from the front** — the discipline comes from *where you add*: stack adds to the front (so newest is popped = LIFO), queue adds to the back (so oldest is polled = FIFO).

```
Deque<Integer> queue = new ArrayDeque<>();

// As STACK (LIFO):    push()/pop()/peek() → operate on FRONT
queue.push(x);        // = offerFirst
queue.pop();           // = pollFirst
queue.peek();          // = peekFirst

// As QUEUE (FIFO):    offer()/poll()/peek() → back-in, front-out
queue.offerLast(x);
queue.pollFirst();
queue.peekFirst();

// As DEQUE:           explicit first/last (e.g. monotone sliding-window)
queue.offerFirst(x);   queue.pollFirst();   queue.peekFirst();
queue.offerLast(x);    queue.pollLast();    queue.peekLast();
```

#503 Next Greater Element II

Description: Given a circular integer array, return the next greater number for every element. The next greater number of an element is the first greater number found by traversing the array circularly.

Variation: Monotone decreasing stack. Iterate through the array TWICE (indices 0 to 2n-1, using `i % n`). Only push index `i` onto the stack during the first pass (`i < n`) to avoid duplicates.

```
class Solution {
    public int[] nextGreaterElements(int[] nums) {
        int n = nums.length;
        int[] result = new int[n];
        Arrays.fill(result, -1);
        Deque<Integer> stack = new ArrayDeque<>(); // monotone decreasing stack of indices
        for (int i = 0; i < 2 * n; i++) {        // ← VARIATION: iterate twice for circular
            while (!stack.isEmpty() && nums[stack.peek()] < nums[i % n])
                result[stack.pop()] = nums[i % n];
            if (i < n) stack.push(i);             // ← VARIATION: only push in first pass
        }
        return result;
    }
}
```

Time O(n) | Space O(n)

#85 Maximal Rectangle

Description: Given a binary matrix filled with 0s and 1s, find the largest rectangle containing only 1s and return its area.

Variation: For each row, compute cumulative histogram heights (1s stack vertically). Then apply the Largest Rectangle in Histogram algorithm (#84) on each row's histogram.

```

class Solution {
    public int maximalRectangle(char[][] matrix) {
        int cols = matrix[0].length;
        int[] heights = new int[cols];
        int maxArea = 0;
        for (char[] row : matrix) {
            for (int j = 0; j < cols; j++)
                heights[j] = row[j] == '1' ? heights[j] + 1 : 0; // ← VARIATION: build histogram row by row
            maxArea = Math.max(maxArea, largestRectangle(heights));
        }
        return maxArea;
    }

    private int largestRectangle(int[] heights) {
        int n = heights.length;
        // Pad with 0s on both ends to flush remaining stack
        int[] h = new int[n + 2];
        System.arraycopy(heights, 0, h, 1, n);
        Deque<Integer> stack = new ArrayDeque<>();
        int max = 0;
        for (int i = 0; i <= n + 1; i++) {
            while (!stack.isEmpty() && h[stack.peek()] > h[i]) {
                int height = h[stack.pop()];
                max = Math.max(max, height * (i - stack.peek() - 1));
            }
            stack.push(i);
        }
        return max;
    }
}

```

Time $O(m \cdot n)$ | **Space** $O(n)$

#224 Basic Calculator

Description: Evaluate a string expression containing `+`, `-`, `(`, `)`, digits, and spaces.

Variation: When encountering `(`, push current (result, sign) onto stack and reset. When encountering `)`, combine current result with the saved result and sign from the stack.

```

class Solution {
    public int calculate(String s) {
        Deque<Integer> stack = new ArrayDeque<>();
        int result = 0, number = 0, sign = 1;
        for (char c : s.toCharArray()) {
            if (Character.isDigit(c)) {
                number = number * 10 + (c - '0');
            } else if (c == '+') {
                result += sign * number; number = 0; sign = 1;
            } else if (c == '-') {
                result += sign * number; number = 0; sign = -1;
            } else if (c == '(') {
                stack.push(result); // ← VARIATION: save result and sign before '('
                stack.push(sign);
                result = 0; sign = 1;
            } else if (c == ')') {
                result += sign * number; number = 0;
                result *= stack.pop(); // ← VARIATION: multiply by sign before '('
                result += stack.pop(); // ← VARIATION: add result before '('
            }
        }
        return result + sign * number;
    }
}

```

Time O(n) | Space O(n)

#227 Basic Calculator II

Description: Evaluate a string expression with `+`, `-`, `*`, `/` and integers (no parentheses).

Variation: Process operator BEFORE the current number. Push positive/negative numbers for `+` `-`. For `*` `/`, immediately multiply/divide the top of the stack. Sum the stack at the end.

```
class Solution {
    public int calculate(String s) {
        Deque<Integer> stack = new ArrayDeque<>();
        int number = 0;
        char sign = '+'; // ← VARIATION: track previous operator
        for (int i = 0; i < s.length(); i++) {
            char c = s.charAt(i);
            if (Character.isDigit(c)) number = number * 10 + (c - '0');
            if ((!Character.isDigit(c) && c != ' ') || i == s.length() - 1) {
                if (sign == '+') stack.push(number);
                else if (sign == '-') stack.push(-number);
                else if (sign == '*') stack.push(stack.pop() * number); // ← VARIATION: apply * immediately
                else if (sign == '/') stack.push(stack.pop() / number); // ← VARIATION: apply / immediately
                sign = c; number = 0;
            }
        }
        return stack.stream().mapToInt(Integer::intValue).sum();
    }
}
```

Time O(n) | Space O(n)

#378 Kth Smallest Element in a Sorted Matrix

Description: Given an $n \times n$ matrix where each row and column is sorted in ascending order, return the k th smallest element.

Variation: Binary search on the VALUE range $[\text{matrix}[0][0], \text{matrix}[n-1][n-1]]$. For a given `mid`, count elements $\leq \text{mid}$ by scanning from the bottom-left corner: if `matrix[row][col] <= mid`, all `row+1` elements in this column (0..row) are $\leq \text{mid}$.

```

class Solution {
    public int kthSmallest(int[][] matrix, int k) {
        int n = matrix.length;
        int i = matrix[0][0], j = matrix[n-1][n-1];          // ← VARIATION: binary search on value range
        while (i < j) {
            int mid = i + (j - i) / 2;
            if (countLE(matrix, mid, n) >= k) {
                j = mid;
            } else {
                i = mid + 1;
            }
        }
        return i;
    }

    private int countLE(int[][] matrix, int target, int n) {
        int count = 0, row = n - 1, col = 0;                // ← VARIATION: scan from bottom-left
        while (row >= 0 && col < n) {
            if (matrix[row][col] <= target) {
                count += row + 1;
                col++;
            } else {
                row--;
            }
        }
        return count;
    }
}

```

Time $O(n \log(\max - \min))$ | **Space** $O(1)$

#373 Find K Pairs with Smallest Sums

Description: Given two sorted arrays `nums1` and `nums2`, return the `k` pairs `(u, v)` (one from each) with the smallest sums.

Variation: Seed min-heap with `(nums1[i], nums2[0])` for `i = 0..min(k, n1.length)-1`. On each pop, if `j+1 < nums2.length`, push `(nums1[i], nums2[j+1])`. This expands row-by-row in the implicit pair matrix.

```

class Solution {
    public List<List<Integer>> kSmallestPairs(int[] nums1, int[] nums2, int k) {
        List<List<Integer>> result = new ArrayList<>();
        if (nums1.length == 0 || nums2.length == 0) return result;
        // heap stores [i, j] meaning pair (nums1[i], nums2[j])
        PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) ->
            nums1[a[0]] + nums2[a[1]] - nums1[b[0]] - nums2[b[1]]);
        for (int i = 0; i < Math.min(nums1.length, k); i++)
            pq.offer(new int[]{i, 0});          // ← VARIATION: seed with (i, 0) for each row
        while (!pq.isEmpty() && result.size() < k) {
            int[] current = pq.poll();
            result.add(Arrays.asList(nums1[current[0]], nums2[current[1]]));
            if (current[1] + 1 < nums2.length)
                pq.offer(new int[]{current[0], current[1] + 1}); // ← VARIATION: expand j in same row
        }
        return result;
    }
}

```

Time $O(k \log k)$ | **Space** $O(k)$

#621 Task Scheduler

Description: Given a list of tasks (characters) and a cooldown `n`, return the minimum number of intervals to finish all tasks. Between two same tasks there must be at least `n` intervals.

Variation: Greedy formula. Find the most frequent task (maxFreq). It creates $\text{maxFreq} - 1$ "frames" of $n + 1$ slots, plus maxCount (count of tasks with maxFreq) slots. The answer is $\max(\text{tasks.length}, (\text{maxFreq} - 1) * (n + 1) + \text{maxCount})$.

```
class Solution {
    public int leastInterval(char[] tasks, int n) {
        int[] count = new int[26];
        for (char c : tasks) count[c - 'A']++;
        Arrays.sort(count);
        int maxFreq = count[25];
        int maxCount = 0;
        for (int c : count) if (c == maxFreq) maxCount++; // ← VARIATION: count ties at max
        return Math.max(tasks.length, (maxFreq - 1) * (n + 1) + maxCount); // ← VARIATION: formula
    }
}
```

Time $O(k)$ | **Space** $O(1)$

#358 Rearrange String k Distance Apart

Description: Rearrange a string so that the same characters are at least k distance apart. Return "" if impossible. **Variation:** greedy with a max-heap by frequency plus a cooldown queue of size k that holds recently used characters until they're eligible again.

```
class Solution {
    public String rearrangeString(String s, int k) {
        if (k <= 1) return s;
        Map<Character, Integer> count = new HashMap<>();
        for (char c : s.toCharArray()) count.merge(c, 1, Integer::sum);
        PriorityQueue<Map.Entry<Character, Integer>> maxHeap =
            new PriorityQueue<>((a, b) -> b.getValue() - a.getValue());
        maxHeap.addAll(count.entrySet());
        Queue<Map.Entry<Character, Integer>> cooldown = new ArrayDeque<>(); // ← VARIATION: cooldown queue
        StringBuilder builder = new StringBuilder();
        while (!maxHeap.isEmpty()) {
            var entry = maxHeap.poll();
            builder.append(entry.getKey());
            entry.setValue(entry.getValue() - 1);
            cooldown.offer(entry);
            if (cooldown.size() < k) continue; // ← VARIATION: wait k slots before reuse
            var ready = cooldown.poll();
            if (ready.getValue() > 0) maxHeap.offer(ready);
        }
        return builder.length() == s.length() ? builder.toString() : "";
    }
}
```

Time $O(n \log k)$ | **Space** $O(k)$

#480 Sliding Window Median

Description: Return the median of every window of size k as it slides across the array. **Variation:** two heaps (maxHeap = lower half, minHeap = upper half) with lazy deletion — defer removing elements that left the window, tracking balance with counts.

```

class Solution {
    public double[] medianSlidingWindow(int[] nums, int k) {
        PriorityQueue<Integer> maxHeap = new PriorityQueue<>((a, b) -> Integer.compare(b, a));
        PriorityQueue<Integer> minHeap = new PriorityQueue<>();
        double[] result = new double[nums.length - k + 1];
        for (int i = 0; i < nums.length; i++) {
            if (maxHeap.isEmpty() || nums[i] <= maxHeap.peek()) {
                maxHeap.offer(nums[i]);
            } else {
                minHeap.offer(nums[i]);
            }
            if (i >= k) {                                     // ← VARIATION: remove element leaving window
                int out = nums[i - k];
                if (out <= maxHeap.peek()) {
                    maxHeap.remove(out);
                } else {
                    minHeap.remove(out);
                }
            }
            // rebalance
            while (maxHeap.size() > minHeap.size() + 1) minHeap.offer(maxHeap.poll());
            while (minHeap.size() > maxHeap.size()) maxHeap.offer(minHeap.poll());
            if (i >= k - 1)
                result[i - k + 1] = (k % 2 == 1) ? (double) maxHeap.peek()
                    : ((double) maxHeap.peek() + minHeap.peek()) / 2.0;
        }
        return result;
    }
}

```

Time $O(n \cdot k)$ with heap remove ($O(n \log k)$ with indexed sets) | **Space** $O(k)$

#692 Top K Frequent Words

Description: Return the k most frequent words. Sort by frequency descending; ties broken by lexicographic order. **Variation:** min-heap of size k ordered so the "smallest" (lowest freq, or same freq with lexicographically larger word) sits on top for eviction.

```

class Solution {
    public List<String> topKFrequent(String[] words, int k) {
        Map<String, Integer> count = new HashMap<>();
        for (String w : words) count.merge(w, 1, Integer::sum);
        PriorityQueue<String> heap = new PriorityQueue<>((a, b) ->
            count.get(a).equals(count.get(b)) ? b.compareTo(a)           // ← VARIATION: larger word evicted first
            : count.get(a) - count.get(b));
        for (String w : count.keySet()) {
            heap.offer(w);
            if (heap.size() > k) heap.poll();
        }
        List<String> result = new ArrayList<>();
        while (!heap.isEmpty()) result.add(heap.poll());
        Collections.reverse(result);
        return result;
    }
}

```

Time $O(n \log k)$ | **Space** $O(n)$

#772 Basic Calculator III

Description: Evaluate an arithmetic expression with `+`, `-`, `*`, `/`, and parentheses. **Variation:** combines #224 (parentheses via recursion) and #227 (operator precedence: apply `*` / `/` immediately, defer `+` / `-`).

```

class Solution {
    private int i;
    public int calculate(String s) { i = 0; return eval(s); }
    private int eval(String s) {
        Deque<Integer> stack = new ArrayDeque<>();
        int num = 0;
        char op = '+';
        while (i < s.length()) {
            char c = s.charAt(i++);
            if (Character.isDigit(c)) num = num * 10 + (c - '0');
            if (c == '(') num = eval(s); // ← VARIATION: recurse on '('
            if ((!Character.isDigit(c) && c != ' ') || i == s.length()) {
                if (op == '+') stack.push(num);
                else if (op == '-') stack.push(-num);
                else if (op == '*') stack.push(stack.pop() * num); // ← VARIATION: precedence
                else if (op == '/') stack.push(stack.pop() / num);
                op = c; num = 0;
            }
            if (c == ')') break; // ← VARIATION: return to caller on ')'
        }
        int sum = 0;
        while (!stack.isEmpty()) sum += stack.pop();
        return sum;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

Additional Reference Problems

#32 Longest Valid Parentheses

Description: Find the length of the longest valid (well-formed) parentheses substring.

Intuition: push indices onto a stack; the bottom of the stack is always the index just before the current valid run, so the gap to it gives the run length.

```

class Solution {
    public int longestValidParentheses(String s) {
        Deque<Integer> stack = new ArrayDeque<>();
        stack.push(-1); // sentinel: index before current valid run
        int result = 0;
        for (int i = 0; i < s.length(); i++) {
            if (s.charAt(i) == '(') {
                stack.push(i);
            } else {
                stack.pop(); // match this ')' with the top '('
                if (stack.isEmpty()) {
                    stack.push(i); // no match: this ')' becomes new base
                } else {
                    result = Math.max(result, i - stack.peek());
                }
            }
        }
        return result;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#71 Simplify Path

Description: Simplify an absolute Unix file path (handle `.`, `..`, multiple slashes).

Intuition: split on `/` and treat directories as a stack — `..` pops the last directory, `.` and empty tokens are ignored.

```
class Solution {
    public String simplifyPath(String path) {
        Deque<String> stack = new ArrayDeque<>();
        for (String part : path.split("/")) {
            if (part.isEmpty() || part.equals(".")) {
                continue;
            } else if (part.equals("..")) {
                if (!stack.isEmpty()) stack.pop(); // go up one directory
            } else {
                stack.push(part);
            }
        }
        StringBuilder builder = new StringBuilder();
        // stack top is the deepest dir; iterate from bottom to build path in order
        Iterator<String> it = stack.descendingIterator();
        while (it.hasNext()) {
            builder.append('/').append(it.next());
        }
        return builder.length() == 0 ? "/" : builder.toString();
    }
}
```

Time $O(n)$ | **Space** $O(n)$

#150 Evaluate Reverse Polish Notation

Description: Evaluate a Reverse Polish Notation expression with `+`, `-`, `*`, `/`.

Intuition: push operands; on an operator pop the two most recent operands, apply, and push the result back — exactly LIFO.

```
class Solution {
    public int evalRPN(String[] tokens) {
        Deque<Integer> stack = new ArrayDeque<>();
        for (String token : tokens) {
            if (token.equals("+") || token.equals("-") || token.equals("*") || token.equals("/")) {
                int b = stack.pop();
                int a = stack.pop(); // order matters for - and /
                if (token.equals("+")) {
                    stack.push(a + b);
                } else if (token.equals("-")) {
                    stack.push(a - b);
                } else if (token.equals("*")) {
                    stack.push(a * b);
                } else {
                    stack.push(a / b);
                }
            } else {
                stack.push(Integer.parseInt(token));
            }
        }
        return stack.pop();
    }
}
```

Time $O(n)$ | **Space** $O(n)$

#215 Kth Largest Element in an Array

Description: Find the kth largest element in an unsorted array (not necessarily distinct).

Intuition: a min-heap of size k keeps the k largest seen so far; its top is the kth largest.

```

class Solution {
    public int findKthLargest(int[] nums, int k) {
        PriorityQueue<Integer> minHeap = new PriorityQueue<>();
        for (int num : nums) {
            minHeap.offer(num);
            if (minHeap.size() > k) minHeap.poll(); // evict smallest, keep k largest
        }
        return minHeap.peek();
    }
}

```

Time $O(n \log k)$ | **Space** $O(k)$

#218 The Skyline Problem

Description: Given building rectangles, return the skyline as a list of key points.

Intuition: sweep left to right over building edges; a max-heap of active heights tracks the tallest standing building, and a key point is emitted whenever that maximum changes.

```

class Solution {
    public List<List<Integer>> getSkyline(int[][] buildings) {
        List<int[]> events = new ArrayList<>();
        for (int[] b : buildings) {
            events.add(new int[]{b[0], -b[2]}); // start: negative height
            events.add(new int[]{b[1], b[2]}); // end: positive height
        }
        // sort by x; at same x, starts before ends, taller start first, shorter end first
        events.sort((a, b) -> a[0] != b[0] ? a[0] - b[0] : a[1] - b[1]);
        List<List<Integer>> result = new ArrayList<>();
        PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
        maxHeap.offer(0); // ground level
        int prevMax = 0;
        for (int[] e : events) {
            if (e[1] < 0) {
                maxHeap.offer(-e[1]); // building starts
            } else {
                maxHeap.remove(e[1]); // building ends
            }
            int currentMax = maxHeap.peek();
            if (currentMax != prevMax) {
                result.add(Arrays.asList(e[0], currentMax));
                prevMax = currentMax;
            }
        }
        return result;
    }
}

```

Time $O(n^2)$ (heap remove) | **Space** $O(n)$

#225 Implement Stack using Queues

Description: Implement a LIFO stack using only queue operations.

Intuition: after each push, rotate the queue so the newest element sits at the front — then `poll` / `peek` behave like stack `pop` / `top` .

```

class MyStack {
    Queue<Integer> queue = new ArrayDeque<>();

    public void push(int x) {
        queue.offer(x);
        for (int i = 1; i < queue.size(); i++) {    // rotate so x moves to front
            queue.offer(queue.poll());
        }
    }
    public int pop()    { return queue.poll(); }
    public int top()    { return queue.peek(); }
    public boolean empty() { return queue.isEmpty(); }
}

```

Time $O(n)$ push, $O(1)$ others | **Space** $O(n)$

#232 Implement Queue using Stacks

Description: Implement a queue using two stacks.

Intuition: one stack receives pushes, the other serves pops; moving elements over reverses their order, restoring FIFO. Amortized $O(1)$ per element.

```

class MyQueue {
    Deque<Integer> inStack = new ArrayDeque<>();    // ← VARIATION: two stacks emulate FIFO
    Deque<Integer> outStack = new ArrayDeque<>();

    public void push(int x) { inStack.push(x); }
    public int pop() {
        peek();                                // ensure outStack is primed
        return outStack.pop();
    }
    public int peek() {
        if (outStack.isEmpty()) {
            while (!inStack.isEmpty()) outStack.push(inStack.pop()); // reverse order
        }
        return outStack.peek();
    }
    public boolean empty() { return inStack.isEmpty() && outStack.isEmpty(); }
}

```

Time $O(1)$ amortized | **Space** $O(n)$

#316 Remove Duplicate Letters

Description: Remove duplicate letters so each appears once, result is lexicographically smallest.

Intuition: monotone increasing stack of letters — pop a larger letter when a smaller one arrives, but only if the popped letter appears again later (so we can re-add it).

```

class Solution {
    public String removeDuplicateLetters(String s) {
        int[] lastIndex = new int[26];
        for (int i = 0; i < s.length(); i++) lastIndex[s.charAt(i) - 'a'] = i;
        boolean[] inStack = new boolean[26];
        Deque<Character> stack = new ArrayDeque<>();
        for (int i = 0; i < s.length(); i++) {
            char c = s.charAt(i);
            if (inStack[c - 'a']) continue;           // keep only one of each letter
            while (!stack.isEmpty() && stack.peek() > c && lastIndex[stack.peek() - 'a'] > i) {
                inStack[stack.pop() - 'a'] = false; // pop bigger letter that recurs later
            }
            stack.push(c);
            inStack[c - 'a'] = true;
        }
        StringBuilder builder = new StringBuilder();
        Iterator<Character> it = stack.descendingIterator();
        while (it.hasNext()) builder.append(it.next());
        return builder.toString();
    }
}

```

Time $O(n)$ | **Space** $O(1)$ (26 letters)

#402 Remove K Digits

Description: Remove k digits from a number string to get the smallest possible number.

Intuition: monotone increasing stack of digits — whenever a smaller digit arrives, pop larger preceding digits (each pop is one removal) so high places hold the smallest digits.

```

class Solution {
    public String removeKdigits(String num, int k) {
        Deque<Character> stack = new ArrayDeque<>();
        for (char c : num.toCharArray()) {
            while (k > 0 && !stack.isEmpty() && stack.peek() > c) {
                stack.pop();           // remove larger leading digit
                k--;
            }
            stack.push(c);
        }
        while (k > 0) {                // still need removals: drop from top
            stack.pop();
            k--;
        }
        StringBuilder builder = new StringBuilder();
        Iterator<Character> it = stack.descendingIterator();
        while (it.hasNext()) builder.append(it.next());
        while (builder.length() > 1 && builder.charAt(0) == '0') builder.deleteCharAt(0);
        return builder.length() == 0 ? "0" : builder.toString();
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#456 132 Pattern

Description: Check if a 1-3-2 pattern exists in an integer array.

Variation: scan right to left with a monotone decreasing stack; pop to track the largest value that is still smaller than some element to its right (the "2"). If a later element is below that "2", a 132 pattern exists.

```

class Solution {
    public boolean find132pattern(int[] nums) {
        Deque<Integer> stack = new ArrayDeque<>(); // candidates for the "3"
        int two = Integer.MIN_VALUE; // ← VARIATION: best valid "2" so far
        for (int i = nums.length - 1; i >= 0; i--) {
            if (nums[i] < two) return true; // nums[i] is the "1" < "2" < "3"
            while (!stack.isEmpty() && stack.peek() < nums[i]) {
                two = stack.pop(); // popped value becomes the "2"
            }
            stack.push(nums[i]);
        }
        return false;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#506 Relative Ranks

Description: Assign ranks ("Gold Medal", "Silver Medal", "Bronze Medal", then 4, 5, ...) to athletes by score.

Intuition: a max-heap of (score, index) pops athletes in descending score order, so each pop assigns the next rank back to its original position.

```

class Solution {
    public String[] findRelativeRanks(int[] score) {
        int n = score.length;
        PriorityQueue<int[]> maxHeap = new PriorityQueue<>((a, b) -> b[0] - a[0]);
        for (int i = 0; i < n; i++) maxHeap.offer(new int[]{score[i], i});
        String[] result = new String[n];
        int rank = 1;
        while (!maxHeap.isEmpty()) {
            int idx = maxHeap.poll()[1]; // highest remaining score
            if (rank == 1) {
                result[idx] = "Gold Medal";
            } else if (rank == 2) {
                result[idx] = "Silver Medal";
            } else if (rank == 3) {
                result[idx] = "Bronze Medal";
            } else {
                result[idx] = String.valueOf(rank);
            }
            rank++;
        }
        return result;
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

#636 Exclusive Time of Functions

Description: Given a log of function start/end times, compute exclusive time per function.

Intuition: a call stack mirrors execution; when a new call starts, the currently running function pauses (accumulate its slice), and when a call ends, the resumed parent restarts its clock.

```

class Solution {
    public int[] exclusiveTime(int n, List<String> logs) {
        int[] result = new int[n];
        Deque<Integer> stack = new ArrayDeque<>();    // function ids currently running
        int prevTime = 0;
        for (String log : logs) {
            String[] parts = log.split(":");
            int id = Integer.parseInt(parts[0]);
            int time = Integer.parseInt(parts[2]);
            if (parts[1].equals("start")) {
                if (!stack.isEmpty()) result[stack.peek()] += time - prevTime; // pause caller
                stack.push(id);
                prevTime = time;
            } else {
                result[stack.pop()] += time - prevTime + 1; // inclusive end timestamp
                prevTime = time + 1;
            }
        }
        return result;
    }
}

```

Time $O(L)$ (L = number of logs) | **Space** $O(n)$

#678 Valid Parenthesis String

Description: Check if a string with `(`, `)`, `*` (wildcard) can be a valid parenthesis string.

Variation: track a range `[low, high]` of possible open-paren counts; `*` widens the range (could be `(`, `)`, or empty). Valid iff the range can return to 0 and never goes negative.

```

class Solution {
    public boolean checkValidString(String s) {
        int low = 0, high = 0;                // ← VARIATION: range of open '(' counts
        for (char c : s.toCharArray()) {
            if (c == '(') {
                low++; high++;
            } else if (c == ')') {
                low--; high--;
            } else {
                // '*': treat as ')' for low, '(' for high
                low--; high++;
            }
            if (high < 0) return false;        // too many ')' even with all '*' as '('
            if (low < 0) low = 0;             // never let open count drop below 0
        }
        return low == 0;
    }
}

```

Time $O(n)$ | **Space** $O(1)$

#703 Kth Largest Element in a Stream

Description: Design a class to find the kth largest element in a stream.

Intuition: a min-heap capped at size k always holds the k largest seen; its top is the running k th largest.

```

class KthLargest {
    private PriorityQueue<Integer> minHeap = new PriorityQueue<>();
    private int k;

    public KthLargest(int k, int[] nums) {
        this.k = k;
        for (int num : nums) add(num);
    }
    public int add(int val) {
        minHeap.offer(val);
        if (minHeap.size() > k) minHeap.poll();    // keep only k largest
        return minHeap.peek();
    }
}

```

Time $O(\log k)$ per add | **Space** $O(k)$

#716 Max Stack

Description: Design a stack with push, pop, top, getMax, and popMax operations.

Variation: mirror an auxiliary `maxStack` (like #155). `popMax` pops down to the max into a temp buffer, removes it, then pushes the buffer back — re-establishing both stacks.

```

class MaxStack {
    Deque<Integer> stack    = new ArrayDeque<>();
    Deque<Integer> maxStack = new ArrayDeque<>();    // ← VARIATION: running max alongside

    public void push(int x) {
        stack.push(x);
        maxStack.push(maxStack.isEmpty() ? x : Math.max(maxStack.peek(), x));
    }
    public int pop() { maxStack.pop(); return stack.pop(); }
    public int top() { return stack.peek(); }
    public int peekMax() { return maxStack.peek(); }
    public int popMax() {
        int max = maxStack.peek();
        Deque<Integer> buffer = new ArrayDeque<>();
        while (top() != max) buffer.push(pop());    // ← VARIATION: unload above the max
        pop();    // remove the max itself
        while (!buffer.isEmpty()) push(buffer.pop()); // reload, rebuilding maxStack
        return max;
    }
}

```

Time $O(n)$ popMax, $O(1)$ others | **Space** $O(n)$

#735 Asteroid Collision

Description: Simulate asteroids moving in a row: right-moving collide with left-moving.

Intuition: a stack holds surviving asteroids; a left-moving asteroid only collides with a right-moving one on top, resolving collisions repeatedly until it survives, explodes, or the stack clears.

```

class Solution {
    public int[] asteroidCollision(int[] asteroids) {
        Deque<Integer> stack = new ArrayDeque<>();
        for (int a : asteroids) {
            boolean alive = true;
            while (alive && a < 0 && !stack.isEmpty() && stack.peek() > 0) {
                if (stack.peek() < -a) {
                    stack.pop();           // top explodes, incoming continues
                } else if (stack.peek() == -a) {
                    stack.pop();           // both explode
                    alive = false;
                } else {
                    alive = false;         // incoming explodes
                }
            }
            if (alive) stack.push(a);
        }
        int[] result = new int[stack.size()];
        for (int i = result.length - 1; i >= 0; i--) result[i] = stack.pop();
        return result;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#759 Employee Free Time

Description: Find the free time intervals common to all employees given their schedules.

Intuition: flatten all intervals, sort by start, then sweep tracking the furthest end seen so far; any gap between that end and the next start is common free time.

```

/*
// Definition for an Interval.
class Interval {
    public int start;
    public int end;
    public Interval() {}
    public Interval(int _start, int _end) { start = _start; end = _end; }
};
*/
class Solution {
    public List<Interval> employeeFreeTime(List<List<Interval>> schedule) {
        List<Interval> all = new ArrayList<>();
        for (List<Interval> emp : schedule) all.addAll(emp);
        all.sort((a, b) -> a.start - b.start);
        List<Interval> result = new ArrayList<>();
        int prevEnd = all.get(0).end;
        for (Interval interval : all) {
            if (interval.start > prevEnd) {
                result.add(new Interval(prevEnd, interval.start)); // gap = free time
            }
            prevEnd = Math.max(prevEnd, interval.end);
        }
        return result;
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

#844 Backspace String Compare

Description: Check if two strings are equal after processing # as backspace.

Intuition: build each string with a stack where `#` pops the last character, then compare the resulting stacks.

```
class Solution {
    public boolean backspaceCompare(String s, String t) {
        return build(s).equals(build(t));
    }
    private String build(String s) {
        Deque<Character> stack = new ArrayDeque<>();
        for (char c : s.toCharArray()) {
            if (c == '#') {
                if (!stack.isEmpty()) stack.pop(); // backspace removes last char
            } else {
                stack.push(c);
            }
        }
        StringBuilder builder = new StringBuilder();
        while (!stack.isEmpty()) builder.append(stack.pop());
        return builder.toString();
    }
}
```

Time $O(n)$ | **Space** $O(n)$

#856 Score of Parentheses

Description: Calculate the score of a valid parentheses string (empty=0, AB=A+B, (A)=2A or 1 if empty).

Variation: stack holds the accumulated score at each depth; push 0 on `(`, and on `)` collapse the inner score (`max(2*inner, 1)`) into the parent frame.

```
class Solution {
    public int scoreOfParentheses(String s) {
        Deque<Integer> stack = new ArrayDeque<>();
        stack.push(0); // ← VARIATION: score of current frame
        for (char c : s.toCharArray()) {
            if (c == '(') {
                stack.push(0); // new inner frame
            } else {
                int inner = stack.pop();
                int add = inner == 0 ? 1 : 2 * inner; // "()"=1, "(A)"=2A
                stack.push(stack.pop() + add); // fold into parent
            }
        }
        return stack.pop();
    }
}
```

Time $O(n)$ | **Space** $O(n)$

#862 Shortest Subarray with Sum at Least K

Description: Find the length of the shortest subarray with sum at least k (k can be negative).

Variation: monotone increasing deque over prefix sums — pop from the front when a window qualifies, and from the back when a newer prefix is smaller (dominating older larger ones).

```

class Solution {
    public int shortestSubarray(int[] nums, int k) {
        int n = nums.length;
        long[] prefix = new long[n + 1];
        for (int i = 0; i < n; i++) prefix[i + 1] = prefix[i] + nums[i];
        Deque<Integer> queue = new ArrayDeque<>(); // indices, prefix increasing front-back
        int result = n + 1;
        for (int i = 0; i <= n; i++) {
            while (!queue.isEmpty() && prefix[i] - prefix[queue.peekFirst()] >= k) {
                result = Math.min(result, i - queue.pollFirst()); // ← VARIATION: window qualifies
            }
            while (!queue.isEmpty() && prefix[queue.peekLast()] >= prefix[i]) {
                queue.pollLast(); // ← VARIATION: drop dominated prefixes
            }
            queue.offerLast(i);
        }
        return result <= n ? result : -1;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#907 Sum of Subarray Minimums

Description: Find the sum of subarray minimums for all subarrays.

Variation: monotone increasing stack — for each element as the minimum, count subarrays via distance to the previous strictly-smaller and next smaller-or-equal element, then weight by the value.

```

class Solution {
    public int sumSubarrayMins(int[] arr) {
        int n = arr.length;
        long MOD = 1_000_000_007L;
        Deque<Integer> stack = new ArrayDeque<>(); // indices, values increasing bottom-top
        long sum = 0;
        for (int i = 0; i <= n; i++) {
            int current = i == n ? Integer.MIN_VALUE : arr[i]; // sentinel flushes stack
            while (!stack.isEmpty() && arr[stack.peek()] > current) {
                int mid = stack.pop(); // arr[mid] is the subarray minimum
                int left = stack.isEmpty() ? -1 : stack.peek();
                long count = (long)(mid - left) * (i - mid); // ← VARIATION: subarrays where mid is min
                sum = (sum + count * arr[mid]) % MOD;
            }
            stack.push(i);
        }
        return (int) sum;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#921 Minimum Add to Make Parentheses Valid

Description: Find the minimum additions to make a parentheses string valid.

Intuition: track open parentheses as a counter (stack of size); each unmatched `)` needs an added `(`, and leftover `(` each need a `)`.

```

class Solution {
    public int minAddToMakeValid(String s) {
        int open = 0; // unmatched '(' (stack height)
        int additions = 0;
        for (char c : s.toCharArray()) {
            if (c == '(') {
                open++;
            } else {
                if (open > 0) {
                    open--; // match an existing '('
                } else {
                    additions++; // need to add a '(' for this ')'
                }
            }
        }
        return additions + open; // remaining '(' each need a ')'
    }
}

```

Time $O(n)$ | **Space** $O(1)$

#946 Validate Stack Sequences

Description: Check if a sequence can be produced by a series of push-pop operations on a stack.

Intuition: simulate — push each value, then greedily pop whenever the stack top matches the next expected popped value. If everything pops, the sequence is valid.

```

class Solution {
    public boolean validateStackSequences(int[] pushed, int[] popped) {
        Deque<Integer> stack = new ArrayDeque<>();
        int j = 0; // index into popped
        for (int value : pushed) {
            stack.push(value);
            while (!stack.isEmpty() && stack.peek() == popped[j]) {
                stack.pop(); // pop as long as top matches
                j++;
            }
        }
        return stack.isEmpty();
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#973 K Closest Points to Origin

Description: Find the k points closest to the origin.

Intuition: a max-heap of size k keyed on squared distance keeps the k closest seen; evict the farthest whenever the heap overflows.

```

class Solution {
    public int[][] kClosest(int[][] points, int k) {
        PriorityQueue<int[]> maxHeap = new PriorityQueue<>(
            (a, b) -> (b[0] * b[0] + b[1] * b[1]) - (a[0] * a[0] + a[1] * a[1]));
        for (int[] point : points) {
            maxHeap.offer(point);
            if (maxHeap.size() > k) maxHeap.poll(); // evict farthest, keep k closest
        }
        int[][] result = new int[k][2];
        for (int i = 0; i < k; i++) result[i] = maxHeap.poll();
        return result;
    }
}

```

Time $O(n \log k)$ | **Space** $O(k)$

#1086 High Five

Description: For each student, compute the average of their top five scores.

Intuition: keep a min-heap of size 5 per student so only their five highest scores remain, then average those.

```

class Solution {
    public int[][] highFive(int[][] items) {
        Map<Integer, PriorityQueue<Integer>> map = new TreeMap<>(); // sorted by student id
        for (int[] item : items) {
            int id = item[0], score = item[1];
            PriorityQueue<Integer> minHeap = map.computeIfAbsent(id, key -> new PriorityQueue<>());
            minHeap.offer(score);
            if (minHeap.size() > 5) minHeap.poll(); // keep top 5 scores
        }
        int[][] result = new int[map.size()][2];
        int i = 0;
        for (Map.Entry<Integer, PriorityQueue<Integer>> e : map.entrySet()) {
            int sum = 0;
            for (int score : e.getValue()) sum += score;
            result[i][0] = e.getKey();
            result[i][1] = sum / 5;
            i++;
        }
        return result;
    }
}

```

Time $O(n \log 5)$ | **Space** $O(n)$

#1190 Reverse Substrings Between Each Pair of Parentheses

Description: Reverse the substrings between each pair of parentheses (innermost first).

Variation: stack of builders — push current builder on (; on) reverse the inner builder and append it to the parent frame.

```

class Solution {
    public String reverseParentheses(String s) {
        Deque<StringBuilder> stack = new ArrayDeque<>(); // ← VARIATION: builder per frame
        StringBuilder current = new StringBuilder();
        for (char c : s.toCharArray()) {
            if (c == '(') {
                stack.push(current); // save parent frame
                current = new StringBuilder();
            } else if (c == ')') {
                current.reverse(); // ← VARIATION: reverse inner substring
                stack.peek().append(current);
                current = stack.pop();
            } else {
                current.append(c);
            }
        }
        return current.toString();
    }
}

```

Time $O(n^2)$ | **Space** $O(n)$

#1209 Remove All Adjacent Duplicates in String II

Description: Remove all adjacent duplicates of length k in a string repeatedly.

Variation: stack of (char, count) pairs — increment the top's count on a repeat, and pop the frame once its count reaches k .

```

class Solution {
    public String removeDuplicates(String s, int k) {
        Deque<int[]> stack = new ArrayDeque<>(); // ← VARIATION: [charCode, runLength]
        for (char c : s.toCharArray()) {
            if (!stack.isEmpty() && stack.peek()[0] == c) {
                stack.peek()[1]++;
                if (stack.peek()[1] == k) stack.pop(); // full group of k: remove
            } else {
                stack.push(new int[]{c, 1});
            }
        }
        StringBuilder builder = new StringBuilder();
        Iterator<int[]> it = stack.descendingIterator();
        while (it.hasNext()) {
            int[] frame = it.next();
            for (int i = 0; i < frame[1]; i++) builder.append((char) frame[0]);
        }
        return builder.toString();
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#1249 Minimum Remove to Make Valid Parentheses

Description: Remove the minimum number of parentheses to make the string valid.

Variation: stack stores indices of unmatched '(' ; a ')' with no match is marked for removal, and any '(' left on the stack at the end is also removed.

```

class Solution {
    public String minRemoveToMakeValid(String s) {
        char[] chars = s.toCharArray();
        Deque<Integer> stack = new ArrayDeque<>();    // ← VARIATION: indices of unmatched '('
        for (int i = 0; i < chars.length; i++) {
            if (chars[i] == '(') {
                stack.push(i);
            } else if (chars[i] == ')') {
                if (stack.isEmpty()) {
                    chars[i] = '*';                // unmatched ')': mark for removal
                } else {
                    stack.pop();
                }
            }
        }
        while (!stack.isEmpty()) chars[stack.pop()] = '*';    // unmatched '('
        StringBuilder builder = new StringBuilder();
        for (char c : chars) {
            if (c != '*') builder.append(c);
        }
        return builder.toString();
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#1337 The K Weakest Rows in a Matrix

Description: Find the k weakest rows in a matrix (fewest soldiers, ties broken by row index).

Variation: max-heap of size k keyed on (soldier count, row index) so the strongest qualifying row sits on top for eviction; the remaining k are the weakest.

```

class Solution {
    public int[] kWeakestRows(int[][] mat, int k) {
        // max-heap: strongest row (more soldiers, or larger index on tie) on top
        PriorityQueue<int[]> maxHeap = new PriorityQueue<>{
            (a, b) -> a[0] != b[0] ? b[0] - a[0] : b[1] - a[1]
        };
        for (int i = 0; i < mat.length; i++) {
            int soldiers = 0;
            for (int v : mat[i]) soldiers += v;
            maxHeap.offer(new int[]{soldiers, i});    // ← VARIATION: (count, index)
            if (maxHeap.size() > k) maxHeap.poll();    // evict strongest
        }
        int[] result = new int[k];
        for (int i = k - 1; i >= 0; i--) result[i] = maxHeap.poll()[1];
        return result;
    }
}

```

Time $O(m \cdot n + m \log k)$ | **Space** $O(k)$

#1541 Minimum Insertions to Balance a Parentheses String

Description: Find the minimum insertions to make a string balanced (every `(` needs two `)`).

Variation: counter-style stack tracking open `(`; each `(` demands two `)`. Handle `)` carefully since they come in pairs, inserting a `(` when only one is available.

```

class Solution {
    public int minInsertions(String s) {
        int open = 0; // unmatched '(' needing ')'
        int insertions = 0;
        int i = 0;
        while (i < s.length()) {
            if (s.charAt(i) == '(') {
                open++;
                i++;
            } else { // ')': need two in a row
                if (i + 1 < s.length() && s.charAt(i + 1) == ')') {
                    i += 2;
                } else {
                    insertions++; // ← VARIATION: insert missing ')'
                    i++;
                }
                if (open > 0) {
                    open--;
                } else {
                    insertions++; // ← VARIATION: insert missing '('
                }
            }
        }
        return insertions + open * 2; // each leftover '(' needs ')'
    }
}

```

Time $O(n)$ | **Space** $O(1)$

#1614 Maximum Nesting Depth of the Parentheses

Description: Find the maximum nesting depth of parentheses in a string.

Intuition: the stack depth equals the current nesting level; track a running counter for `( / )` and record its peak.

```

class Solution {
    public int maxDepth(String s) {
        int depth = 0; // current stack height
        int result = 0;
        for (char c : s.toCharArray()) {
            if (c == '(') {
                depth++;
                result = Math.max(result, depth);
            } else if (c == ')') {
                depth--;
            }
        }
        return result;
    }
}

```

Time $O(n)$ | **Space** $O(1)$

#1792 Maximum Average Pass Ratio

Description: Maximize the average pass ratio by adding extra students optimally.

Variation: max-heap keyed on the marginal gain from adding one student to a class; greedily assign each extra student where it helps most, then push the updated gain back.

```

class Solution {
    public double maxAverageRatio(int[][] classes, int extraStudents) {
        // max-heap by gain from adding one passing student
        PriorityQueue<double[]> maxHeap = new PriorityQueue<>(
            (a, b) -> Double.compare(b[2], a[2]));    // ← VARIATION: order by marginal gain
        for (int[] c : classes) {
            double pass = c[0], total = c[1];
            maxHeap.offer(new double[]{pass, total, gain(pass, total)});
        }
        while (extraStudents-- > 0) {
            double[] top = maxHeap.poll();
            double pass = top[0] + 1, total = top[1] + 1; // add one passing student
            maxHeap.offer(new double[]{pass, total, gain(pass, total)});
        }
        double sum = 0;
        for (double[] c : maxHeap) sum += c[0] / c[1];
        return sum / classes.length;
    }
    private double gain(double pass, double total) {
        return (pass + 1) / (total + 1) - pass / total;
    }
}

```

Time $O((n + \text{extra}) \log n)$ | **Space** $O(n)$

#1944 Number of Visible People in a Queue

Description: Count the number of people each person can see in a queue (taller blocks view).

Variation: monotone decreasing stack scanned right to left; pop shorter people (each is visible) until a taller one blocks the view — that taller person is visible too.

```

class Solution {
    public int[] canSeePersonsCount(int[] heights) {
        int n = heights.length;
        int[] result = new int[n];
        Deque<Integer> stack = new ArrayDeque<>();    // heights, decreasing bottom-top
        for (int i = n - 1; i >= 0; i--) {
            while (!stack.isEmpty() && stack.peek() < heights[i]) {
                stack.pop();                        // ← VARIATION: shorter person is visible
                result[i]++;
            }
            if (!stack.isEmpty()) result[i]++;        // the taller blocker is also visible
            stack.push(heights[i]);
        }
        return result;
    }
}

```

Time $O(n)$ | **Space** $O(n)$

#1985 Find the Kth Largest Integer in the Array

Description: Find the kth largest integer in an array of number strings.

Variation: min-heap of size k comparing the numeric strings by length first, then lexicographically (so big-integer values compare correctly without overflow).

```
class Solution {
    public String kthLargestNumber(String[] nums, int k) {
        // ← VARIATION: compare by length then lexicographically (numeric order for big ints)
        PriorityQueue<String> minHeap = new PriorityQueue<>(
            (a, b) -> a.length() != b.length() ? a.length() - b.length() : a.compareTo(b));
        for (String num : nums) {
            minHeap.offer(num);
            if (minHeap.size() > k) minHeap.poll(); // keep k largest
        }
        return minHeap.peek();
    }
}
```

Time $O(n \log k)$ | **Space** $O(k)$